

Theory

R is a high-level programming language used for statistical computing, data analysis, and graphical representation. This practical covers two core aspects of R:

1. File Import and Export:

- `read.table()` / `write.table()` — for tab-separated .txt files.
- `read.csv()` / `write.csv()` — for comma-separated .csv files.
- `readLines()` — reads raw content of a file line by line.

2. Loop Control Structures:

- for loop — iterates over a sequence or vector; number of iterations known in advance.
- while loop — repeats as long as a condition is TRUE; exits when condition fails.
- repeat loop — executes indefinitely; exit is controlled explicitly using `break`.
- next — skips the current iteration and moves to the next.
- break — immediately exits the loop regardless of condition.

Part I — Import and Export: Text File and CSV File

Code for Program 1a — Create Data Frame and Export to TXT File:

```
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1a_export_txt.R
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1 # Program 1a: Create student data frame and export to TXT
2
3 # Step 1: Create data frame
4 student_data <- data.frame(
5   RollNo = c(101, 102, 103, 104, 105, 106),
6   Name   = c("Aarav Sharma", "Priya Desai", "Rahul Patil",
7             "Sneha Joshi", "Amit Verma", "Neha Kulkarni"),
8   Age    = c(21, 22, 20, 23, 21, 22),
9   Marks  = c(88, 76, 92, 65, 81, 57),
10  Grade  = c("A", "B", "A", "C", "B", "D"),
11  City   = c("Nagpur", "Mumbai", "Pune",
12            "Nagpur", "Delhi", "Aurangabad")
13 )
14
15 # Step 2: Display the data frame
16 cat("--- Student Dataset ---\n")
17 print(student_data)
18
19 # Step 3: Export to TXT
20 write.table(student_data,
21             file      ="students.txt",
22             sep       ="\\t",
23             row.names =FALSE,
24             quote     =FALSE)
25 cat("\nExported to students.txt\n")
```

Output for Program 1a:

```
Console ~/
> student_data
  RollNo      Name Age Marks Grade      City
1    101    Aarav Sharma  21   88     A    Nagpur
2    102    Priya Desai  22   76     B    Mumbai
3    103    Rahul Patil  20   92     A     Pune
4    104    Sneha Joshi  23   65     C    Nagpur
5    105    Amit Verma  21   81     B     Delhi
6    106    Neha Kulkarni 22   57     D Aurangabad
> cat("\nExported to students.txt\n")

Exported to students.txt
```

Code for Program 1b — Import from Text File and Field Operations:

```
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1b_import_txt.R
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1 # Program 1b: Import from TXT and perform field operations
2
3 # Step 1: Import
4 imported_txt <- read.table("students.txt",
5                           header = TRUE,
6                           sep = "\t")
7
8 # Step 2: Display
9 cat("--- Imported from students.txt ---\n")
10 print(imported_txt)
11
12 # Step 3: Field operations
13 cat("\n-- Field Operations --\n")
14 cat("Total Students :", nrow(imported_txt), "\n")
15 cat("Average Marks :", mean(imported_txt$Marks), "\n")
16 cat("Highest Marks :", max(imported_txt$Marks), "\n")
17 cat("Lowest Marks :", min(imported_txt$Marks), "\n")
18
19 # Step 4: Filter Marks >= 80
20 cat("\n-- Students with Marks >= 80 --\n")
21 print(imported_txt[imported_txt$Marks >= 80, ])
22
23 # Step 5: Filter by City
24 cat("\n-- Students from Nagpur --\n")
25 print(imported_txt[imported_txt$City == "Nagpur", ])
```

Output for Program 1b:

```
Console ~/

> cat("--- Imported from students.txt ---\n")
--- Imported from students.txt ---
> print(imported_txt)
  RollNo      Name Age Marks Grade      City
1    101  Aarav Sharma  21   88    A   Nagpur
2    102   Priya Desai  22   76    B   Mumbai
3    103   Rahul Patil  20   92    A    Pune
4    104   Sneha Joshi  23   65    C   Nagpur
5    105   Amit Verma  21   81    B    Delhi
6    106  Neha Kulkarni  22   57    D Aurangabad

-- Field Operations --
Total Students   : 6
Average Marks    : 76.5
Highest Marks     : 92
Lowest Marks      : 57

-- Students with Marks >= 80 --
  RollNo      Name Age Marks Grade      City
1    101 Aarav Sharma  21   88    A   Nagpur
3    103 Rahul Patil  20   92    A    Pune
5    105 Amit Verma  21   81    B    Delhi

-- Students from Nagpur --
  RollNo      Name Age Marks Grade      City
1    101 Aarav Sharma  21   88    A   Nagpur
4    104 Sneha Joshi  23   65    C   Nagpur
```

Code for Program 1c — Export to CSV File:

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1c_export_csv.R

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```
1 # Program 1c: Export student data to CSV file
2
3 write.csv(student_data,
4           file         = "students.csv",
5           row.names    = FALSE)
6
7 cat("Exported to students.csv successfully.\n")
8
9 # Verify: preview raw CSV content
10 cat("\n-- CSV File Preview --\n")
11 cat(readLines("students.csv"), sep = "\n")
```

Output for Program 1c:

```
Console ~/

> cat("Exported to students.csv successfully.\n")
Exported to students.csv successfully.
> cat(readLines("students.csv"), sep = "\n")
"RollNo","Name","Age","Marks","Grade","City"
"101","Aarav Sharma","21","88","A","Nagpur"
"102","Priya Desai","22","76","B","Mumbai"
"103","Rahul Patil","20","92","A","Pune"
"104","Sneha Joshi","23","65","C","Nagpur"
"105","Amit Verma","21","81","B","Delhi"
"106","Neha Kulkarni","22","57","D","Aurangabad"
```

Code for Program 1d — Import from CSV and Analysis:

```
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1d_import_csv.R
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1 # Program 1d: Import from CSV and perform analysis
2
3 imported_csv <- read.csv("students.csv")
4
5 cat("--- Imported from students.csv ---\n")
6 str(imported_csv)
7 print(imported_csv)
8
9 cat("\n-- Summary: Age and Marks --\n")
10 print(summary(imported_csv[, c("Age", "Marks")]))
11
12 cat("\n-- Grade-wise Student Count --\n")
13 print(table(imported_csv$Grade))
14
15 grade_A <- imported_csv[imported_csv$Grade == "A", ]
16 write.csv(grade_A, file = "grade_A_students.csv", row.names = FALSE)
17 cat("\nGrade A students exported to grade_A_students.csv\n")
```

Output for Program 1d:

```
Console ~/
> str(imported_csv)
data.frame': 6 obs. of 6 variables:
 $ RollNo: int  101 102 103 104 105 106
 $ Name  : chr  "Aarav Sharma" "Priya Desai" "Rahul Patil" ...
 $ Age   : int  21 22 20 23 21 22
 $ Marks : int  88 76 92 65 81 57
 $ Grade : chr  "A" "B" "A" "C" ...
 $ City  : chr  "Nagpur" "Mumbai" "Pune" "Nagpur" ...

-- Summary: Age and Marks --
      Age      Marks
Min.   :20.00  Min.   :57.00
1st Qu.:21.00  1st Qu.:67.75
Median :21.50  Median :78.50
Mean   :21.50  Mean   :76.50
3rd Qu.:22.00  3rd Qu.:85.75
Max.   :23.00  Max.   :92.00

-- Grade-wise Student Count --
A B C D
2 2 1 1

Grade A students exported to grade_A_students.csv
```

Part II — Loop Operations: for, while, repeat

Code for Program 2a — for Loop:

```
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2a_for_loop.R
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1 # Program 2a: for Loop demonstrations
2
3 # 1. Basic for loop: 1 to 5
4 cat("--- for loop: 1 to 5 ---\n")
5 for (i in 1:5) {
6   cat("Iteration:", i, "\n")
7 }
8
9 # 2. Iterate over character vector
10 cat("\n--- for loop: Student Names ---\n")
11 names_vec <- c("Aarav", "Priya", "Rahul", "Sneha", "Amit", "Neha")
12 for (name in names_vec) {
13   cat("Student:", name, "\n")
14 }
15
16 # 3. Accumulate sum of marks
17 cat("\n--- for loop: Sum of Marks ---\n")
18 marks <- c(88, 76, 92, 65, 81, 57)
19 total <- 0
20 for (m in marks) {
21   total <- total + m
22 }
23 cat("Total Marks:", total, "\n")
24 cat("Average   :", total / length(marks), "\n")
25
26 # 4. next and break
27 cat("\n--- for: Odd numbers, stop at 7 ---\n")
28 for (i in 1:10) {
29   if (i %% 2 == 0) next
30   if (i > 7) break
31   cat("Odd:", i, "\n")
32 }
33
34 # 5. Grade classification
35 cat("\n--- for: Grade Classification ---\n")
36 marks2 <- c(88, 76, 92, 65, 81, 57)
37 names2 <- c("Aarav", "Priya", "Rahul", "Sneha", "Amit", "Neha")
38 for (i in 1:length(marks2)) {
39   grade <- ifelse(marks2[i] >= 90, "A+",
40                 ifelse(marks2[i] >= 80, "A",
41                       ifelse(marks2[i] >= 70, "B", "C")))
42   cat(names2[i], ":", marks2[i], "-> Grade:", grade, "\n")
43 }
```

Output for Program 2a — for Loop:

```
Console ~/
> cat("--- for loop: 1 to 5 ---\n")
--- for loop: 1 to 5 ---
Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
Iteration: 5

--- for loop: Student Names ---
Student: Aarav   Student: Priya   Student: Rahul
Student: Sneha   Student: Amit    Student: Neha

--- for loop: Sum of Marks ---
Total Marks: 459
Average      : 76.5

--- for: Odd numbers, stop at 7 ---
Odd: 1   Odd: 3   Odd: 5   Odd: 7

--- for: Grade Classification ---
Aarav : 88 -> Grade: A+   Priya : 76 -> Grade: B
Rahul : 92 -> Grade: A+   Sneha : 65 -> Grade: C
Amit  : 81 -> Grade: A    Neha  : 57 -> Grade: C
```

Code for Program 2b — while Loop:

```
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2b_while_loop.R
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1 # Program 2b: while Loop demonstrations
2
3 # 1. Countdown
4 cat("--- while loop: Countdown ---\n")
5 count <- 5
6 while (count > 0) {
7   cat("Count:", count, "\n")
8   count <- count - 1
9 }
10 cat("Countdown complete!\n")
11
12 # 2. Running total until > 250
13 cat("\n--- while: Running Total ---\n")
14 marks <- c(88, 76, 92, 65, 81, 57)
15 total <- 0; idx <- 1
16 while (total <= 250 && idx <= length(marks)) {
17   total <- total + marks[idx]
18   cat("Added marks[", idx, "] =", marks[idx], "| Total:", total, "\n")
19   idx <- idx + 1
20 }
21 cat("Stopped. Total =", total, "\n")
22
23 # 3. Fibonacci up to 100
24 cat("\n--- while: Fibonacci up to 100 ---\n")
25 a <- 0; b <- 1
26 cat(a, b)
27 while ((a + b) <= 100) {
28   temp <- a + b
29   cat(" ", temp)
30   a <- b; b <- temp
31 }
32 cat("\n")
33
34 # 4. Find first topper (Marks > 85)
35 cat("\n--- while: First Topper ---\n")
36 names3 <- c("Aarav", "Priya", "Rahul", "Sneha", "Amit", "Neha")
37 marks3 <- c(88, 76, 92, 65, 81, 57)
38 i <- 1
39 while (i <= length(marks3)) {
40   if (marks3[i] > 85) {
41     cat("First topper:", names3[i], "with", marks3[i], "marks\n")
42     break
43   }
44   i <- i + 1
45 }
```

Output for Program 2b — while Loop:

```
Console ~/
> cat("--- while loop: Countdown ---\n")
--- while loop: Countdown ---
Count: 5
Count: 4
Count: 3
Count: 2
Count: 1
Countdown complete!

--- while: Running Total ---
Added marks[ 1 ] = 88 | Total: 88
Added marks[ 2 ] = 76 | Total: 164
Added marks[ 3 ] = 92 | Total: 256
Stopped. Total = 256

--- while: Fibonacci up to 100 ---
0 1 1 2 3 5 8 13 21 34 55 89

--- while: First Topper ---
First topper: Aarav with 88 marks
```

Code for Program 2c — repeat Loop:

```
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2c_repeat_loop.R
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1 # Program 2c: repeat Loop demonstrations
2
3 # 1. Print 1 to 5
4 cat("--- repeat loop: 1 to 5 ---\n")
5 i <- 1
6 repeat {
7   cat("Value:", i, "\n")
8   i <- i + 1
9   if (i > 5) break
10 }
11
12 # 2. Below-average students
13 cat("\n--- repeat: Below Average Students ---\n")
14 marks4 <- c(88, 76, 92, 65, 81, 57)
15 names4 <- c("Aarav", "Priya", "Rahul",
16            "Sneha", "Amit", "Neha")
17 avg <- mean(marks4)
18 cat("Class Average:", avg, "\n")
19 j <- 1
20 repeat {
21   if (marks4[j] < avg)
22     cat("Below Average:", names4[j], "->", marks4[j], "\n")
23   j <- j + 1
24   if (j > length(marks4)) break
25 }
26
27 # 3. Sentinel value: sum until 0
28 cat("\n--- repeat: Sum until 0 ---\n")
29 inputs <- c(10, 20, 35, 0)
30 total2 <- 0; k <- 1
31 repeat {
32   val <- inputs[k]
33   if (val == 0) {
34     cat("Exit. Final Sum:", total2, "\n")
35     break
36   }
37   total2 <- total2 + val
38   cat("Added:", val, "| Running Sum:", total2, "\n")
39   k <- k + 1
40 }
```

Output for Program 2c — repeat Loop:

```
Console ~/
> cat("--- repeat loop: 1 to 5 ---\n")
--- repeat loop: 1 to 5 ---
Value: 1
Value: 2
Value: 3
Value: 4
Value: 5

--- repeat: Below Average Students ---
Class Average: 76.5
Below Average: Priya -> 76
Below Average: Sneha -> 65
Below Average: Neha -> 57

--- repeat: Sum until 0 ---
Added: 10 | Running Sum: 10
Added: 20 | Running Sum: 30
Added: 35 | Running Sum: 65
Exit. Final Sum: 65
```

Conclusion

Thus, the following R programs were successfully written, executed in RStudio, and the outputs were verified:

- Program 1a: A 6-record student dataset (RollNo, Name, Age, Marks, Grade, City) was created as an R data frame and exported to students.txt using write.table() with tab separator.
- Program 1b: The text file was re-imported using read.table() and field-level operations (total count, average, max, min, filtering by marks and city) were performed successfully.
- Program 1c: The same student data was exported to students.csv using write.csv() and the raw CSV content was verified using readLines().
- Program 1d: The CSV was imported using read.csv(), summary statistics and grade-wise counts were generated, and a filtered subset (Grade A students) was exported to a new file.
- Program 2a (for loop): Demonstrated sequence iteration, vector traversal, mark accumulation, next/break usage, and grade classification using ifelse().
- Program 2b (while loop): Demonstrated countdown, running total with threshold exit, Fibonacci series generation up to 100, and finding the first student above a mark threshold.
- Program 2c (repeat loop): Demonstrated fixed-count iteration with break, identifying below-average students, and sentinel-value based sum accumulation.

All programs were executed in RStudio and produced the expected outputs. File import/export operations and all three loop types in R were successfully demonstrated.